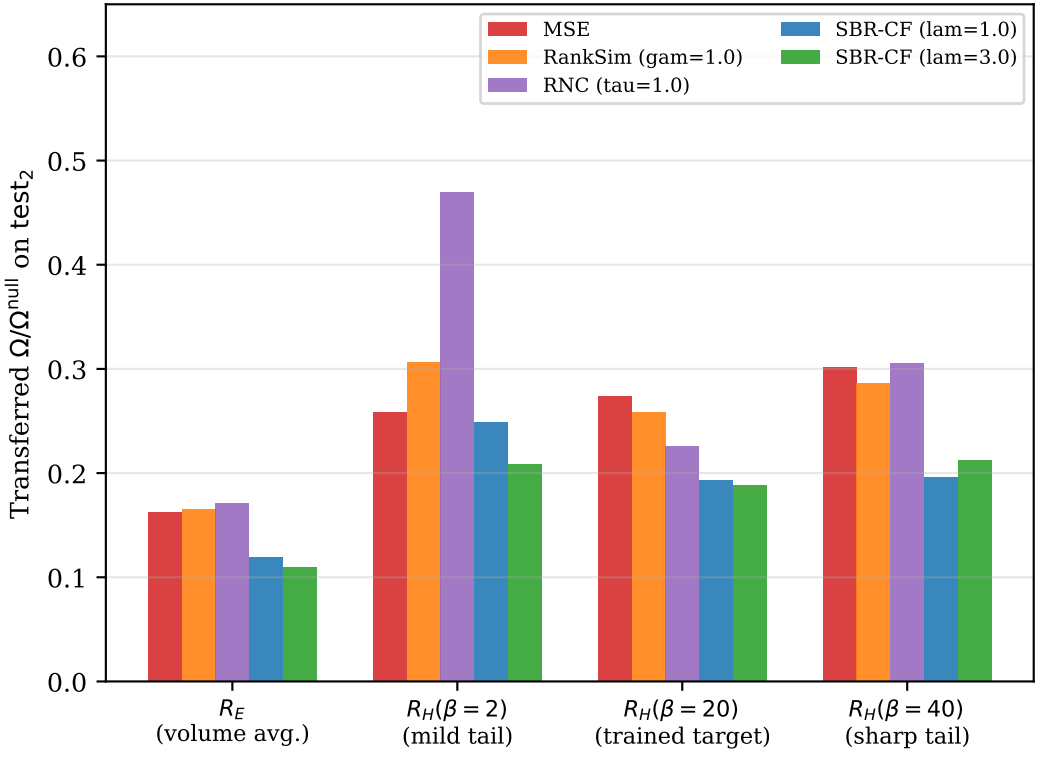
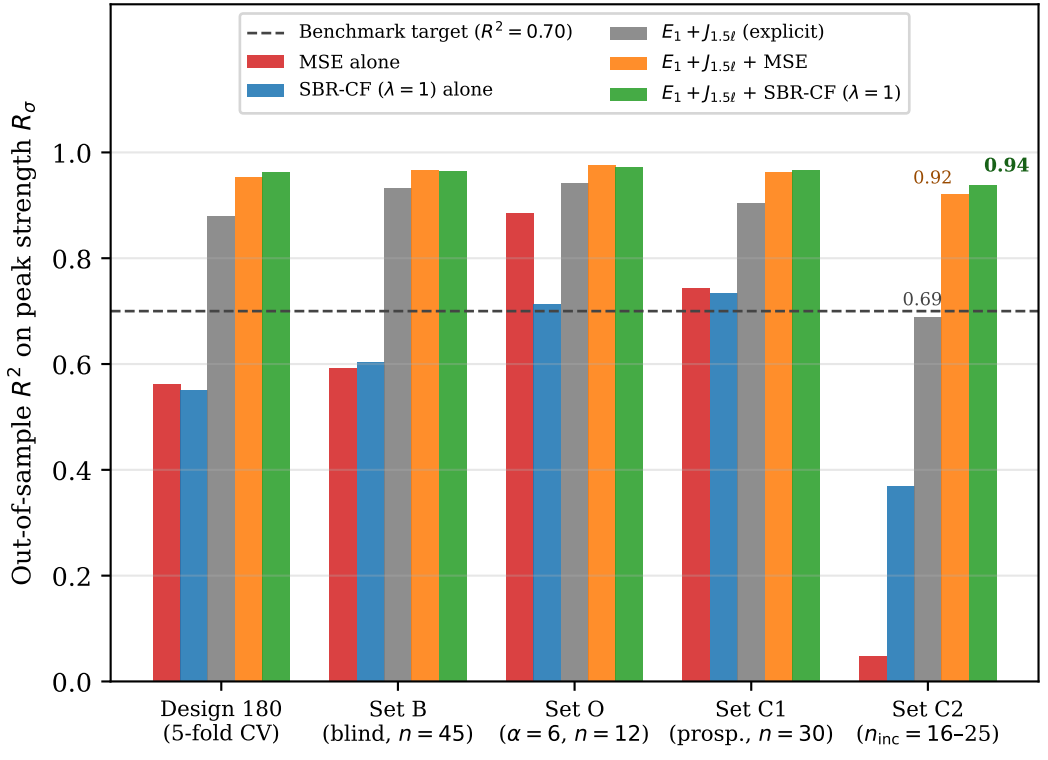

(a) Cross-horizon transfer of $\beta = 20$ encoders
MSE
RankSim (gam=1.0)
RNC (tau=1.0)
SBR-CF (lam=1.0)
SBR-CF (lam=3.0)
Transferred $\Omega/\Omega^{\text{null}}$ on test_2
0.0
0.1
0.2
0.3
0.4
0.5
0.6
R_E (volume avg.)
$R_H(\beta = 2)$ (mild tail)
$R_H(\beta = 20)$ (trained target)
$R_H(\beta = 40)$ (sharp tail)


(b) Frozen transfer to AT1 phase-field fracture (R_σ)
Benchmark target ($R^2 = 0.70$)
MSE alone
SBR-CF ($\lambda = 1$) alone
$E_1 + J_{1.5\ell}$ (explicit)
$E_1 + J_{1.5\ell}$ + MSE
$E_1 + J_{1.5\ell}$ + SBR-CF ($\lambda = 1$)
Out-of-sample R^2 on peak strength R_σ
0.0
0.2
0.4
0.6
0.8
1.0
0.69
0.92
0.94
Design 180 (5-fold CV)
Set B (blind, $n = 45$)
Set O ($\alpha = 6$, $n = 12$)
Set C1 (prosp., $n = 30$)
Set C2 ($n_{\text{inc}} = 16$–25)